away from metaphysics and theology." In this, they succeeded for a time. Philosophers like Hans Reichenbach in Germany (who had his own Berlin Circle of philosophers) and A. J. Ayer in England visited Vienna, then returned home and promoted logical positivism across national borders and language barriers. Rudolf Carnap became the leading exponent of the Vienna Circle's views; his landmark 1929 book *The Logical Structure of the World* established him as a towering figure in the positivist movement, and many of his students went on to become important philosophers in their own right. Carnap and Reichenbach managed to take over an existing philosophy journal, *Annalen der Philosophie*, and turned it to their own ends, renaming it *Erkenntnis* ("knowledge" or "realization") and publishing articles on positivism from their own circles and elsewhere. Meanwhile, Otto Neurath, ebullient and enormous by nature—"as untidy and rumbustious as Schlick was elegant and urbane," Ayer recalled, "a giant of a man who used to sign his letters with a drawing of an elephant"—worked on several ambitious schemes to change the world in the name of the unity of science. He started a grand encyclopedia project, the *International Encyclopedia of Unified Science*, meant to explain the ideas of positivism and the sciences in one authoritative multivolume reference. He worked to develop an international symbolic language, ISOTYPE, that would precisely specify sense data in unambiguous ways, to aid international collaboration in science and philosophy. And Neurath also organized a series of conferences—International Congresses for Unified Science—where positivists from around the world met and discussed the progress of their philosophical and social program. For a brief moment in the late 1920s and early 1930s, the promise of the Vienna Circle's manifesto burned bright.

Many of the positivists' ideas—their emphasis on observation, their dismissal of "reality" and unseen entities as metaphysics, and their idea of science as a mere instrument for organizing perceptions—sound similar to some of the ideas associated with the Copenhagen interpretation. Logical positivism and quantum physics came out of the same time and place: the